

[Report Title]

Team Members: [Zihan Zhang, Xiuqi Cao, Junyu Zhou]

Project: [News Source Classification]

1 Introduction

A brief description (i.e., less than 1 page) of the highlights of the project (e.g., model design, performance, findings from either core or exploratory questions).

2 Core Components

In this section, we describe our approach to data preprocessing, model design, and evaluation. We first outline how the dataset is prepared and split. We then present our model design and iterative improvements. Finally, we describe our evaluation protocol and metrics used to compare model performance.

2.1 Data Preprocessing

[ZZ: Need an example graph: row data – > processed data]

We construct our dataset by processing a collection of news article URLs and extracting structured inputs suitable for text classification. Each URL is first parsed to derive a pseudo-headline, which serves as the textual input for our models, while the corresponding label is inferred from the source domain.

Specifically, we begin by reading the dataset from a CSV file containing article URLs. For each URL, we perform a series of normalization and cleaning steps. We first remove query parameters and fragment identifiers to obtain a clean URL. The URL is then split into path segments, and the final meaningful segment is selected as the basis for constructing the headline. In cases where the last segment contains non-informative identifiers (e.g., record IDs such as “rcrd”), we instead use the preceding segment.

We further clean the extracted segment by removing file-specific suffixes (e.g., “.print”) and filtering out dataset-specific artifacts such as trailing article IDs commonly found in NBC URLs. The remaining text is transformed into a pseudo-headline by replacing hyphens with spaces, removing non-alphanumeric characters, converting all text to lowercase, and collapsing redundant whitespace.

Finally, we assign labels based on the source domain of the URL. Articles from `foxnews.com` are labeled as FoxNews, while all others are labeled as NBC. This results in a dataset of cleaned text inputs paired with binary labels, which is then used for downstream model training and evaluation.

2.2 Two-Stage Model Exploration and Refinement

In this project, we adopt a two-stage modeling strategy to systematically explore and refine different approaches for short-text classification. Rather than relying on a single model, our goal is to identify effective representations and architectures through a structured exploration process, followed by targeted refinement of the most promising models.

[XC: Check the following model]

First stage, we perform a coarse-grained exploration across different representation paradigms. For each representation type, we further instantiate multiple model architectures to capture different inductive biases, enabling a joint comparison of representation and model design choices.

- **Sparse representations:** TF-IDF features combined with linear classifiers, specifically Logistic Regression and Support Vector Machines (SVM).

1 简介

项目亮点的简要说明 (即少于 1 页), 如模型设计、性能、核心或探索性问题的发现。

2 核心组成部分

在本节中, 我们将介绍我们对数据预处理、模型设计和评估的方法。我们首先概述数据集的准备和拆分方式。随后我们展示了模型设计和迭代改进。最后, 我们介绍了我们的评估方案和用于比较模型性能的指标。

2.1 数据预处理

[ZZ: 需要示例图: 行数据 — 处理数据>]

我们通过处理一组新闻文章 URL 并提取适合文本分类的结构化输入来构建数据集。每个 URL 首先被解析以推导出伪标题, 作为模型的文本输入, 而相应的标签则从源域推断。

具体来说, 我们首先从包含文章 URL 的 CSV 文件中读取数据集。对于每个 URL, 我们会执行一系列归一化和清理步骤。我们首先移除查询参数和片段标识符, 以获得干净的 URL。然后将 URL 拆分为路径段, 最终选择有意义的段作为构建标题的基础。如果最后一段包含非信息标识符 (例如记录 ID, 如“rcrd”), 我们会使用前一段。

我们进一步清理提取段, 去除文件特定的后缀 (如“.print”), 并过滤掉数据集特有的伪影, 如常见于 NBC 网址的尾部文章 ID。剩余文本通过用空格替换连字符、去除非字母数字字符、将所有文本转换为小写字母以及折叠冗余空白, 转化为伪标题。

最后, 我们根据 URL 的源域名分配标签。foxnews.com 的文章标注为福克斯新闻, 其他所有文章标注为 NBC。这会产生一个由清理后文本输入与二进制标签匹配的数据集, 随后用于下游模型训练和评估。

2.2 两阶段模型探索与精炼

本项目采用两阶段建模策略, 系统地探索和完善不同的短文本分类方法。我们的目标不是依赖单一模型, 而是通过结构化的探索过程识别有效的表示和架构, 随后针对最有前景的模型进行精炼。

[XC: 请查看以下型号]

第一阶段, 我们对不同的表示范式进行粗粒度探索。对于每种表示类型, 我们还进一步实例化了多种模型架构, 以捕捉不同的归纳偏见, 从而实现表示与模型设计选择的联合比较。

- 稀疏表示: TF-IDF 特征结合线性分类器, 特别是逻辑回归和支持向量机 (SVM)。

- **Static embeddings:** Pretrained word embeddings (GloVe) combined with neural models, including a Convolutional Neural Network (CNN) and a Multi-Layer Perceptron (MLP).
- **Contextual embeddings:** Transformer-based models, including BERT and RoBERTa, which are fine-tuned for the classification task.

Within each category, we experiment with multiple model variants to capture different inductive biases. This stage allows us to compare fundamentally different approaches and identify the best-performing model within each representation family.

Second stage, we focus on iterative refinement of the selected models. Specifically, we take the best model from each representation group and further improve its performance through targeted modifications and hyperparameter tuning. This includes exploring architectural variations, training strategies, and key design choices specific to each model family. By narrowing our focus to the most promising candidates, we are able to perform more fine-grained optimization without incurring excessive computational cost.

This two-stage framework enables us to balance breadth and depth in model exploration. The first stage provides a principled comparison across diverse representation methods, while the second stage allows for detailed optimization within each family. As a result, we are able to isolate the impact of representation and architecture, and develop a deeper understanding of which modeling choices are most effective for news headline classification.

[JZ: Need an overall graph]

2.3 Model Evaluation

In this section, we evaluate model performance on the processed dataset described in Section 2.1, using the two-stage training procedure introduced in Section 2.2.

2.3.1 Experimental Configurations

We consider six models across three representation paradigms. For each model, we define the input representation x , model function $f_\theta(\cdot)$, predicted probability $\hat{y}$, and the corresponding loss function $\mathcal{L}$.

TF-IDF + Logistic Regression Given a text input x , we extract a TF-IDF feature vector $\mathbf{v} \in \mathbb{R}^d$. The model is defined as

$$\hat{y} = \sigma(\mathbf{w}^\top \mathbf{v} + b),$$

where $\mathbf{w} \in \mathbb{R}^d$, $b \in \mathbb{R}$, and $\sigma(\cdot)$ is the sigmoid function. The loss is binary cross-entropy:

$$\mathcal{L} = -(y \log \hat{y} + (1 - y) \log(1 - \hat{y})).$$

Hyperparameters include TF-IDF dimension d and regularization strength λ .

TF-IDF + SVM Using the same TF-IDF feature $\mathbf{v} \in \mathbb{R}^d$, the model predicts

$$\hat{y} = \text{sign}(\mathbf{w}^\top \mathbf{v} + b).$$

The loss is hinge loss:

$$\mathcal{L} = \max(0, 1 - y(\mathbf{w}^\top \mathbf{v} + b)).$$

Hyperparameter is the regularization strength λ .

GloVe + MLP Let a sequence of tokens be represented as embeddings $\{\mathbf{e}_1, \dots, \mathbf{e}_T\}$, where $\mathbf{e}_i \in \mathbb{R}^k$. We compute the average embedding

$$\mathbf{h}_0 = \frac{1}{T} \sum_{i=1}^T \mathbf{e}_i.$$

This is passed through a multi-layer perceptron:

$$\mathbf{h}_1 = \phi(\mathbf{W}_1 \mathbf{h}_0 + \mathbf{b}_1), \quad \hat{y} = \sigma(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2),$$

- 静态嵌入：预训练词嵌入（GloVe）结合神经模型，包括卷积神经网络（CNN）和多层感知器（MLP）。
- 上下文嵌入：基于变换器的模型，包括 BERT 和 RoBERTa，针对分类任务进行微调。

在每个类别中，我们尝试多种模型变体以捕捉不同的归纳偏差。这一阶段使我们能够比较根本不同的方法，并识别每个表示族中表现最佳的模型。

第二阶段，我们专注于对所选模型进行迭代优化。具体来说，我们从每个表示组中选出最佳模型，通过有针对性修改和超参数调优进一步提升其性能。这包括探索架构变体、训练策略以及各模型家族特有的关键设计选择。通过将关注点缩小到最有潜力的候选者，我们能够进行更细致的优化，同时避免过高的计算成本。

这一两阶段框架使我们在模型探索中能够平衡广度与深度。第一阶段提供对不同表示方法的原则性比较，第二阶段则允许在每个族内进行详细优化。因此，我们能够分离出表示方式和架构的影响，并深入理解哪些建模选择最适合新闻标题分类。

[JZ：需要整体图表]

2.3 模型评估

本节采用第 2.2 节引入的两阶段训练程序，评估模型在第 2.1 节描述的处理数据集上的表现。

2.3.1 实验配置

我们考虑了三种表示范式中的六个模型。对于每个模型，我们定义输入表示 x ，模型函数 $f(\cdot)$ ，预测概率 $\hat{y}$ ，以及对应的损失函数 L 。

TF-IDF + 逻辑回归给定文本输入 x ，我们提取一个 TF-IDF 特征向量 $v \in \mathbb{R}$ 。该模型定义为

$y = \sigma(wv + b)$ ，其中 $w \in \mathbb{R}$ ， $b \in \mathbb{R}$ ， $\sigma(\cdot)$ 是 S 形函数。损失为二元交叉熵：

$$L = -y \log \hat{y} - (1 - y) \log (1 - \hat{y}).$$

超参数包括 TF-IDF 维数 d 和正则化强度 λ 。

TF-IDF + SVM 利用相同的 TF-IDF 特征 $v \in \mathbb{R}$ ，模型预测

$$\hat{y} = \text{符号}(wv + b).$$

损失是铰链损失：

$$L = \max(0, 1 - y(wv + b)).$$

超参数是正则化强度 λ 。

GloVe + MLP 令一组令牌用嵌入 $\{e_1, \dots, e_T\}$ 表示，其中 $e_i \in \mathbb{R}$ 。我们计算平均嵌入

$$h = \frac{1}{T} \sum_{i=1}^T e_i.$$

该信号通过多层感知器：

$$h = \phi(Wh + b), \quad \hat{y} = \sigma(Wh + b),$$

Model	Key Parameters	Accuracy	F1-score
TF-IDF + Logistic Regression			
TF-IDF + SVM			
GloVe + MLP			
GloVe + CNN			
BERT			
RoBERTa			

Table 1: Stage 1: Performance comparison across different models

where $\phi(\cdot)$ is ReLU. The loss is cross-entropy. Hyperparameters include embedding dimension k and hidden layer number and size.

GloVe + CNN Given embeddings $\mathbf{E} \in \mathbb{R}^{T \times k}$, we apply 1D convolutions:

$$\mathbf{c}_j = \text{ReLU}(\mathbf{W}_j * \mathbf{E} + b_j),$$

followed by max-pooling:

$$\mathbf{h} = \max_j \mathbf{c}_j.$$

The pooled features are passed to a linear layer:

$$\hat{y} = \sigma(\mathbf{w}^\top \mathbf{h} + b).$$

The loss is cross-entropy. Parameters include kernel sizes, number of filters, and embedding dimension k .

BERT Given tokenized input x , BERT produces contextual representations:

$$\mathbf{h} = \text{BERT}(x).$$

Then the prediction is estimated as:

$$\hat{y} = \text{softmax}(\mathbf{W}\mathbf{h} + \mathbf{b}).$$

The loss is cross-entropy. Hyperparameters include learning rate η , batch size, and number of epochs.

RoBERTa Similar to BERT, we obtain contextual representation:

$$\mathbf{h} = \text{RoBERTa}(x),$$

and compute:

$$\hat{y} = \text{softmax}(\mathbf{W}\mathbf{h} + \mathbf{b}).$$

The loss is cross-entropy. Hyperparameters include learning rate η , batch size, and training epochs.

Metrics. We evaluate model performance based on the confusion matrix, where TP (true positives), FP (false positives), TN (true negatives), and FN (false negatives) summarize prediction outcomes, with total samples $n = TP + TN + FP + FN$. Based on these quantities, we report overall performance metrics including **accuracy** $\frac{TP+TN}{n}$ and prediction quality metrics **F1-score** $\frac{2TP}{2TP+FP+FN}$, which balances precision and recall. Together, these metrics provide a comprehensive evaluation of model performance.

Data Split: The dataset is divided into training, validation, and test sets using a fixed split ratio: 70% training, 10% validation, and 20% testing. The same split is used across all models to ensure comparability.

Implementations. We implement all experiments in Python. The experiments are conducted on a server equipped with a 2.4 GHz CPU and 0.75 TB of RAM, with access to two A100 GPUs, each with 40 GB of memory.

模型	关键参数	精度	F1 评分	TF-IDF + 逻辑回归	TF-IDF + SVM	GloVe + MLP
GloVe + CNN						
伯特 罗伯塔						

表 1：第一阶段：不同型号的性能比较

其中 $\phi(\cdot)$ 是 ReLU。损失是交叉熵。超参数包括嵌入维数 k 以及隐藏层数和大小。

GloVe + CNN 给定嵌入 $E \in \mathbb{R}$ ，我们应用一维卷积：

$$c = \text{ReLU}(W * E + b),$$

随后进行最大池化：

$$h = \max_j c_j.$$

合并后的特征会传递到线性层：

$$\hat{y} = \sigma(wh + b).$$

损失是交叉熵。参数包括核大小、滤波器数量和嵌入维数 k 。

BERT 给定分词输入 x ，BERT 生成上下文表示：

$$h = \text{BERT}(x).$$

然后预测估计为：

$$\hat{y} = \text{softmax}(Wh + b).$$

损失是交叉熵。超参数包括学习率 η 、批次大小和纪元数。

RoBERTa 与 BERT 类似，我们得到上下文表示：

$$h = \text{RoBERTa}(x),$$

并计算：

$$\hat{y} = \text{softmax}(Wh + b).$$

损失是交叉熵。超参数包括学习率 η 、批次大小和训练历时。

指标。我们基于混淆矩阵评估模型表现，其中 TP（真阳性）、FP（假阳性）、TN（真阴性）和 FN（假阴性）汇总预测结果，总样本 $n = TP + TN + FP + FN$ 。基于这些数据，我们报告包括准确率 n 和预测质量指标 F1-score 2 在内的整体性能指标，平衡了准确性和召回率。这些指标共同提供了模型性能的全面评估。

数据分配：数据集按固定比例分为训练、验证和测试集：70%训练，10%验证，20%测试。所有模型采用相同的分配以确保可比性。

实现。我们所有实验都用 Python 实现。实验在一台配备 2.4 GHz CPU 和 0.75 TB 内存的服务器上进行，并可访问两块 A100 GPU，每块 40 GB 内存。

Some Parameter	Accuracy	F1-score

Table 2: **Model 1** Effect of regularization parameter X

Some Parameter	Accuracy	F1-score

Table 3: **Model 2** Effect of regularization parameter X

2.3.2 Stage 1: Broad Model Exploration

In the first stage, we perform a broad exploration across different representation paradigms and model architectures. For each model, we use a fixed set of baseline hyperparameters and evaluate performance on the validation set. The goal of this stage is to identify the most promising models within each representation family.

[XC: Describe the performance here.]

2.3.3 Stage 2: Fine-Grained Model Refinement

In the second stage, we perform fine-grained hyperparameter tuning on the best-performing models from each representation family. Specifically, we explore the impact of key design choices and training parameters to further improve performance.

[XC: Describe the performance here.]

2.3.4

3 Exploratory Components

In this section, we explore extensions beyond the baseline to compare different modeling approaches and analyze how text representation affects performance.

3.1 Technique: Structured Model Exploration for Text Classification

We adopt a structured approach to model exploration, focusing on understanding the role of representation in short-text classification. Rather than treating model selection as a purely empirical process, we organize our exploration along different representation paradigms and systematically compare their effectiveness.

The core idea is to decouple representation and model design, allowing us to analyze how different levels of text representation—ranging from sparse features to contextual embeddings—affect classification performance. This structured perspective provides insights into the trade-offs between model complexity and representation quality.

Detailed implementation and experimental results are presented in Section 2.2 and Section 2.3.

3.2 Analysis: Understanding the Role of Representation in Short Text Classification

To better understand the impact of representation on model performance, we investigate the following research questions:

某些参数精度 F1 评分		

表 2：模型 1 正则化参数 X 的影响

某些参数精度 F1 评分		

表 3：模型 2 正则化参数 X 的影响

2.3.2 第一阶段：广义模型探索

第一阶段，我们对不同的表示范式和模型架构进行了广泛的探索。对于每个模型，我们使用固定的基线超参数集，并评估验证集的性能。这一阶段的目标是识别每个表示族中最有前景的模型。

[XC：请在这里描述表演。]

2.3.3 第二阶段：细粒度模型精细化

第二阶段，我们对每个表现族中表现最好的模型进行细粒度超参数调优。具体来说，我们探讨了关键设计选择和训练参数对进一步提升性能的影响。

[XC：请在这里描述表演。]

2.3.4

3 探索性组成部分

本节将探讨基线之外的扩展，比较不同的建模方法，并分析文本表示如何影响性能。

3.1 技术：文本分类的结构化模型探索

我们采用结构化方法进行模型探索，重点理解表征在短文本分类中的作用。我们不将模型选择视为纯粹的实证过程，而是沿着不同的表示范式组织探索，并系统地比较其有效性。核心思想是将表示法与模型设计解耦，使我们能够分析不同层级的文本表示——从稀疏特征到上下文嵌入——如何影响分类性能。这种结构化视角为模型复杂度与表示质量之间的权衡提供了见解。

详细的实现和实验结果见第 2.2 节和第 2.3 节。

3.2 分析：理解表征在短文本分类中的作用

为了更好地理解表示对模型性能的影响，我们探讨了以下研究问题：

Some Parameter	Accuracy	F1-score

Table 4: **Model 3** Effect of regularization parameter X

- Do contextual embedding models significantly outperform traditional methods such as TF-IDF in short-text classification?
- Are simpler architectures (e.g., CNN) sufficient to capture meaningful patterns in short texts such as news headlines?

Based on these questions, we formulate the following hypotheses:

- **H1:** Contextual embedding models (e.g., BERT) outperform traditional methods due to their ability to capture richer semantic and contextual information.
- **H2:** For short texts, local n-gram features (captured by CNN) can be competitive with more complex sequence-based models.

Based on our experimental results, [\[XC: discuss using the results\]](#).

4 Team Contributions

Zihan Zhang participated in data collection, data preprocessing, and model design; Xiuqi Cao participated in model design and training the models; and junyu Zhou participated in model design and writing the report.

某些参数精度 F1 评分	

表 4：模型 3 正则化参数 X 的影响

- 上下文嵌入模型在短文本分类中是否显著优于传统方法如 TF-IDF?
- 更简单的架构（例如 CNN）是否足以在新闻标题等短文本中捕捉有意义的模式？

我们调查以下研究问题：基于这些问题，我们提出了以下假设：

- H1：上下文嵌入模型（如 BERT）因其能够捕捉更丰富的语义和上下文信息，优于传统方法。
- H2：对于短文本，局部 n-gram 特征（由 CNN 捕获）可以与更复杂的基于序列的模型竞争。

根据我们的实验结果，[XC：讨论使用结果]。

4 团队贡献

张志涵参与了数据收集、数据预处理和模型设计;曹秀奇参与了模型设计和模型的训练;周君宇参与模型设计和报告撰写。